MATH3900 September 29, 2026
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Assignment 2
Due: 11:59 p.m., October 20, 2026

Instructions: *100 pts*

1. Give precise and adequate explanations to support your conclusions. Clarity of presentation counts. You may cite results proved in class without reproving them.
2. Discussions are encouraged. Plagiarism is not allowed. Prepare your own work.
3. Submit one PDF on Gradescope. LATE SUBMISSIONS WILL NOT BE ACCEPTED.
4. Append the AI-use declaration specified at the end, including if you used no AI. **Omitting it results in a deduction of 50 points from the assignment mark.**

Throughout, $\|\cdot\|$ denotes the Euclidean norm for vectors and its induced operator norm for matrices.

1. For each statement, determine whether it is true or false. If true, provide a proof; if false, provide a counterexample and verify it.
   (a) Let $f:\mathbb{R}^d\to\mathbb{R}$ be twice continuously differentiable and suppose $\nabla^2 f(x)$ is invertible. Replacing f by $af+b$, where $a>0$ and $b\in\mathbb{R}$, leaves the Newton direction at x unchanged. [4]
   (b) Let $f:\mathbb{R}^d\to\mathbb{R}$ be continuously differentiable. If $\alpha_\star>0$ minimizes $f(x+\alpha p)$ over $\alpha\geq 0$, then $\nabla f(x+\alpha_\star p)^T p=0$. [4]
   (c) Fix a continuously differentiable objective, a point x, and a descent direction p. Armijo backtracking with fixed parameters $\bar{\alpha}>0$ and $c_{\mathrm{A}},\eta\in(0,1)$ produces the same next point if p is replaced by sp, for any $s>0$. [4]
   (d) Let $f:\mathbb{R}^d\to\mathbb{R}$ be bounded below and L-smooth. If every gradient descent update $x_{k+1}=x_k-\alpha_k\nabla f(x_k)$ satisfies the Armijo condition with the same $c_{\mathrm{A}}\in(0,1)$ and some $\alpha_k>0$, then $\|\nabla f(x_k)\|\to 0$. [4]
   (e) Let $f:\mathbb{R}\to\mathbb{R}$ be twice continuously differentiable, with $f''(t)>0$ for every $t\in\mathbb{R}$, and suppose f has a global minimizer $t^\star$. If one full Newton step reaches $t^\star$ from every starting point $t_0\in\mathbb{R}$, then f is a quadratic function. [4]
2. Consider $f(t)=\sqrt{1+t^2}$ on $\mathbb{R}$.
   (a) Compute $f'(t)$ and $f''(t)$, find the global minimizer, and derive the full Newton update as a function of the current iterate t_k. [6]

(b) Starting from $t_0 = 2$, perform two Newton iterations with Armijo backtracking, using $\bar{\alpha} = 1$, $\eta = 1/2$, and $c_{\mathrm{A}} = 1/4$ at each iteration. For every trial tested, give the step size, the trial objective value, and the Armijo bound. State the accepted iterates t_1, t_2. [8]

(c) For the full Newton iteration, determine all starting points $t_0 \in \mathbb{R}$ for which t_k converges to the minimizer. Describe its behavior for every other starting point. For $0 < |t_0| < 1$, find $q > 1$ such that $\lim_{k\to\infty} |t_{k+1}|/|t_k|^q$ is finite and strictly positive, and compute this limit. [6]

3. Let

$$f(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2 + \frac{1}{4}(x_1 + x_2)^4, \qquad x^{(0)} = (1, 0)^T.$$

(a) Compute $\nabla f(x^{(0)})$ and $\nabla^2 f(x^{(0)})$, and calculate one full Newton step $x^{(1)}$. [6]

(b) Introduce coordinates $u = x_1 + x_2$ and $v = x_1 - x_2$, and define $F(u, v) = f((u + v)/2, (u - v)/2)$. Write F explicitly. Starting from the point corresponding to $x^{(0)}$, calculate one full Newton step for F, then express the resulting point in the original x coordinates. [8]

(c) Let $f : \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable, let $A \in \mathbb{R}^{d\times d}$ be invertible, and let $b \in \mathbb{R}^d$. Define $F(z) = f(Az + b)$. At corresponding points $x = Az + b$, assume $\nabla^2 f(x)$ is invertible. Prove that the full Newton updates for f and F satisfy $x^+ = Az^+ + b$. [6]

4. Let $f : \mathbb{R}^d \to \mathbb{R}$ be bounded below and L-smooth, with $L > 0$. At each nonstationary iterate, choose $p_k \in \mathbb{R}^d$ satisfying

$$-\nabla f(x_k)^T p_k \geq a\|\nabla f(x_k)\|\,\|p_k\|, \qquad \|p_k\| \geq b\|\nabla f(x_k)\|, \tag{1}$$

where $a \in (0, 1]$ and $b > 0$ are fixed. Apply the algorithm *Descent method with backtracking line search* from the lecture slides, using $\varepsilon = 0$ and fixed parameters $\bar{\alpha} > 0$ and $c_{\mathrm{A}}, \eta \in (0, 1)$. If a stationary point is reached, keep all subsequent iterates equal to it.

(a) Find a constant $\delta > 0$, depending only on $a, b, L, \bar{\alpha}, c_{\mathrm{A}}, \eta$, such that $\|x_{k+1} - x_k\| \geq \delta\|\nabla f(x_k)\|$ for every k. Prove your bound. [6]

(b) Find a constant $C > 0$, depending only on $a, b, L, \bar{\alpha}, c_{\mathrm{A}}, \eta$, such that $f(x_{k+1}) \leq f(x_k) - C\|\nabla f(x_k)\|^2$ for every k. Prove your bound. [4]

(c) Assume in addition that f is μ-strongly convex, where $0 < \mu \leq L$, and let $x^\star$ be its minimizer. Find $q \in (0, 1)$ in terms of μ and your constant C from part (b) such that, for every integer $k \geq 0$, [6]

$$f(x_k) - f(x^\star) \leq q^k\big(f(x_0) - f(x^\star)\big).$$

Prove your bound and deduce that $x_k \to x^\star$.

(d) Suppose instead that each nonstationary iterate uses exact line search: choose $\alpha_k \in \operatorname{argmin}_{\alpha \geq 0} f(x_k + \alpha p_k)$, assuming this minimum is attained. Prove that the conclusion of part (b) remains valid with the same C, and that the conclusions of part (c) hold with the same q under its additional strong-convexity assumption. [4]

5. An *inexact Newton method* uses an approximate solution of the Newton equation. Define its residual by

$$r_k = \nabla^2 f(x_k) p_k + \nabla f(x_k).$$

(a) Let $f(x) = x_1^2/2 + 2x_2^2$, $x = (1, 1/4)^T$, and $r = (5/4, 1/4)^T$. Solve $\nabla^2 f(x) p = -\nabla f(x) + r$. Check whether $\|r\| < \|\nabla f(x)\|$, and determine whether p is a descent direction. [4]

For parts (b)–(d), let $f : \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable, μ-strongly convex, and L-smooth, where $0 < \mu \leq L$. At every nonstationary iterate require

$$\|r_k\| \leq \theta \|\nabla f(x_k)\|, \qquad 0 \leq \theta < \mu/L,$$

with fixed θ. Use the Armijo backtracking rule of Question 4, and keep the iterates fixed if a stationary point is reached.

(b) Find constants $a \in (0, 1]$ and $b > 0$ for which the directions satisfy both conditions in (1). Justify your choices. [6]

(c) Apply Question 4 to prove that the inexact Newton iterates converge to the unique minimizer $x^\star$ from every initial point. Give a geometric bound on $f(x_k) - f(x^\star)$ in terms of k, the parameters, and $f(x_0) - f(x^\star)$. [6]

(d) Suppose the following additional assumptions hold. [4]

(i) The Hessian $\nabla^2 f$ is Lipschitz near $x^\star$.
(ii) The accepted step sizes satisfy $\alpha_k = 1$ for all sufficiently large k.
(iii) The residuals satisfy $\|r_k\| \leq D \|\nabla f(x_k)\|^2$ for some fixed $D > 0$.

Prove that, for all sufficiently large k, $\|x_{k+1} - x^\star\| \leq K \|x_k - x^\star\|^2$ for some constant $K > 0$.

AI-use declaration (required). Append a short statement using the highest level that describes your AI use on any part:

0. No AI tools used.
1. Limited: brainstorming, concept explanations, or language editing.
2. Substantial: help developing a proof, calculation, or solution.
3. Extensive: AI generated most of the initial draft of an answer.

Use this format, naming the affected parts and how you checked the assistance:

AI use: Level [0/1/2/3]. Tool(s): [names, or "none"]. Parts and purpose: [brief description, or "none"]. Checks: [how you verified the mathematics, or "not applicable"]. I take responsibility for my submitted work.

A Level 0 declaration is required if you used no AI. **Omitting the declaration results in a deduction of 50 points from the homework mark.**